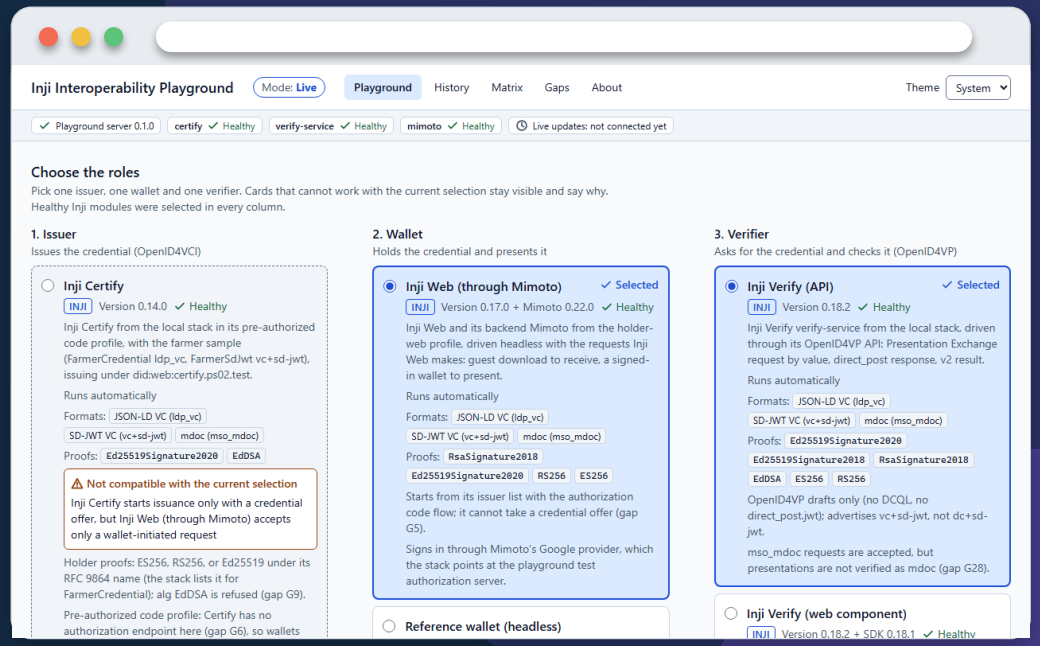
MOSIP DECODE 2026 · PROBLEM STATEMENT 02

Inji Interoperability Playground

Pick an issuer, a wallet and a verifier, run a real credential flow, and read every protocol message



Team Badmosip

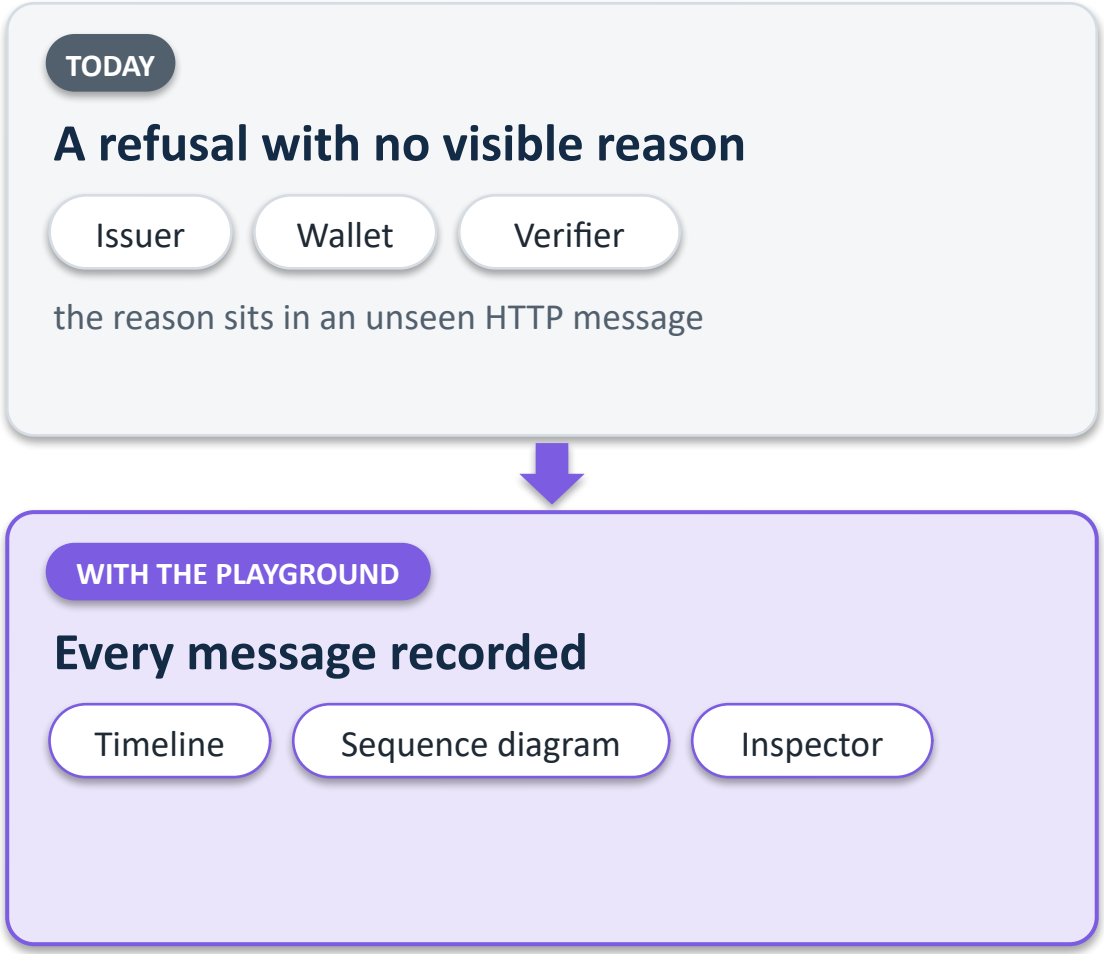
Kanav Bhardwaj · Ivan Bhargava · Nikunj Mahajan
Ishaan Sodhi · Harsh Modi

Nobody can see whether Inji modules work together

1. **Inji supports the standards**
   OpenID4VCI, OpenID4VP, W3C VC, SD-JWT VC

2. **No tool runs them end to end**
   across Inji modules, or against code from outside MOSIP

3. **Failures hide in the traffic**
   the reason for a refusal sits in an unseen HTTP message



TARGET AUDIENCE

Made for people who must trust Inji across vendors



New integrators

evaluating Inji before they commit



Inji core teams

tracking regressions across modules and versions



National ID programmes

comparing wallets and verifiers from different vendors

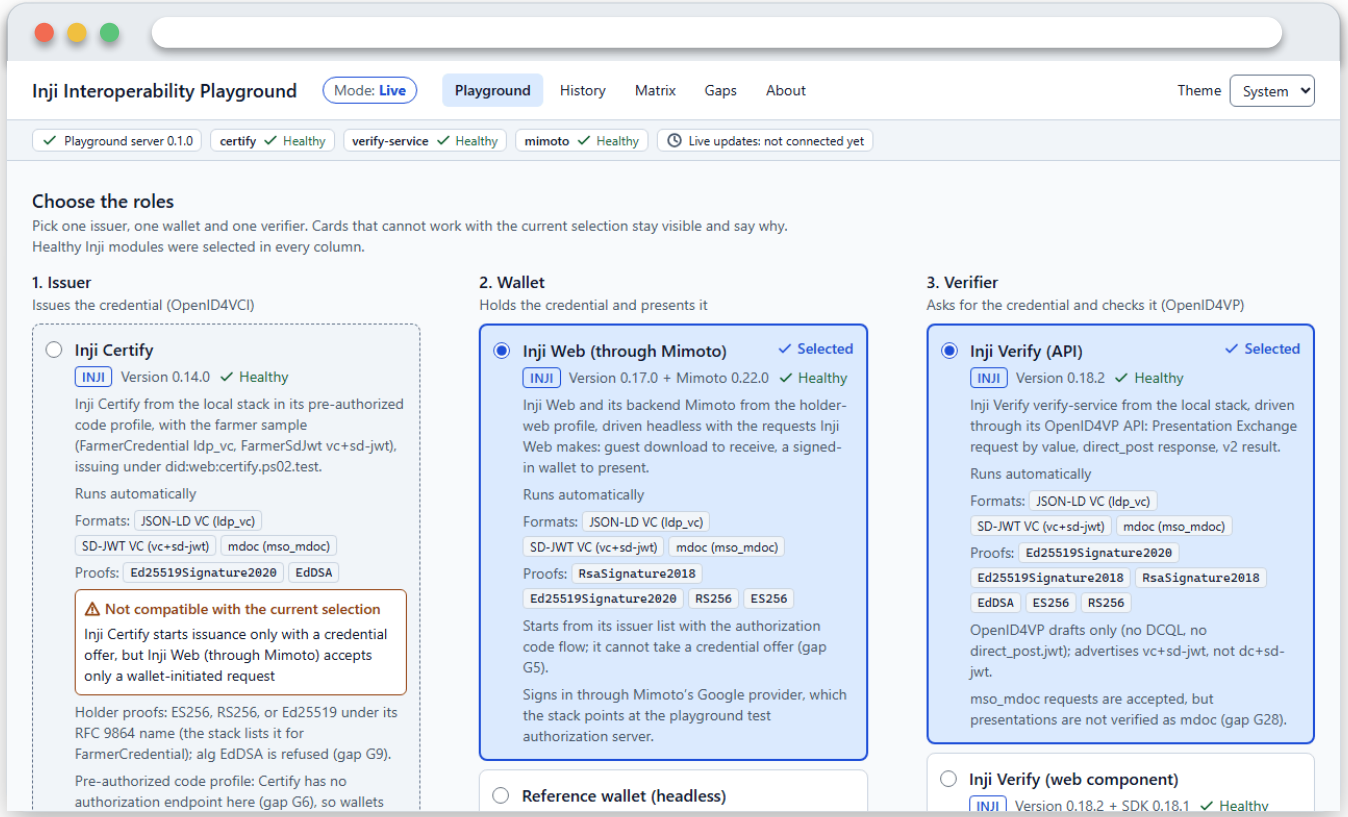
Pick three roles, run a real flow, read every message

1. **Three columns**
   issuer, wallet, verifier: Inji or reference roles

2. **Every message recorded**
   timeline, sequence diagram, inspector

3. **21 scenarios**
   happy path, 14 tamper cases, 6 probes

4. **4 formats, 2 protocol profiles**
   draft and 1.0 OpenID4VCI and OpenID4VP



Choosing the roles; cards that cannot work together say why

ARCHITECTURE

Roles are the unit of interoperability

Browser UI: React, Vite, Tailwind
playground, run page, inspector, matrix, history, gaps

REST /api, WebSocket /ws

Playground server (Fastify, Node 24)

Orchestrator
role registry, compatibility check

Reference roles
issuer, wallet, verifier from the specs

Inji adapters
Certify, Inji Web, Verify, Verify SDK

Capture points
outbound, inbound, reverse proxy

SQLite store
reports, matrix, gaps

HTTP through the capture points

Local Inji stack (Docker Compose)

Inji Certify 0.14.0
pre-authorized and authorization code

Mimoto 0.22.0
with Inji Web 0.17.0

Inji Verify 0.18.2
service and UI

TLS front
*.ps02.test, test CA

TECHNOLOGY STACK

TypeScript from server to browser, run against real Inji images

01 Languages and runtime

- TypeScript
- Node.js 24
- SQLite (node:sqlite)

02 Frameworks and tools

- Fastify server, zod schemas
- React 18, Vite, Tailwind
- Vitest, Playwright with axe checks

03 MOSIP and Inji

- Inji Certify 0.14.0
- Mimoto 0.22.0 and Inji Web 0.17.0
- Inji Verify 0.18.2 and react-inji-verify-sdk

04 Infrastructure and standards

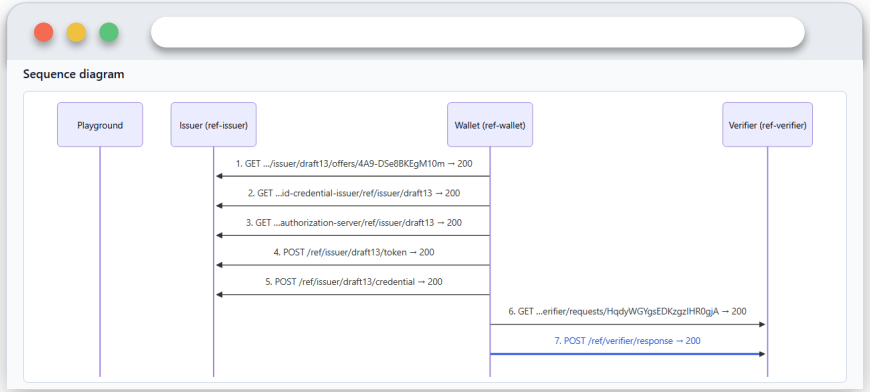
- Docker Compose, PostgreSQL 15, TLS front
- GitHub Actions: ci and interop-nightly
- OpenID4VCI, OpenID4VP, SD-JWT VC, mdoc

DEMO

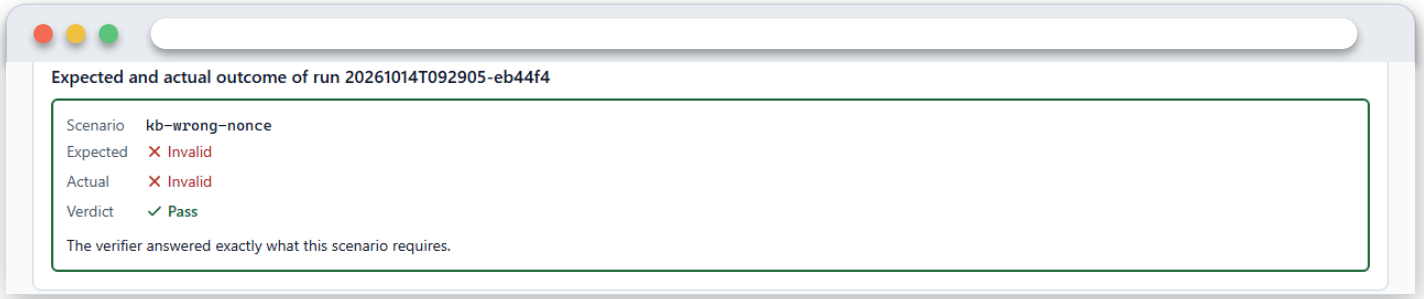
A wrong nonce must be refused, and the report proves it

1. **Start**
   npm run demo prints DEMO PASS
2. **Choose**
   issuer, wallet, verifier, format
3. **Watch**
   timeline and sequence diagram fill in
4. **Report**
   expected and actual outcome
5. **Tamper**
   a wrong nonce must be refused
6. **Matrix and gaps**
   every cell links to its run

Demo video:
https://youtu.be/Bbgxy05Z8MM



Sequence diagram: each arrow is one recorded HTTP message



Tamper case: expected Invalid, actual Invalid, verdict Pass

EVIDENCE

1253 of 1253 tests pass, and every live cell comes out as specified

1253

unit and integration tests

1253 of 1253 passed

20 / 20

live tests

against the Inji stack, 129 s

123 / 123

reference matrix cells

as specified, no Docker

Also measured: 12 / 12 browser tests (with axe accessibility checks) · 5 / 5 demo cells (DEMO PASS, 249 s with --ci) · 16 / 16 live-core cells (as specified against Inji) · 10 / 10 live Inji Web cells (as specified) · 2 green hosted CI workflows (ci and interop-nightly) · 'As specified' means each cell gave the verdict its spec expects; an expected fail is a recorded Inji gap.

REQUIREMENTS COVERED

Every mandatory task is done, each with a screenshot or run record

Mandatory tasks

- ✓ **M1 Inji stack end to end**
  interop-report/live/matrix.md
- ✓ **M2 Three-column selector**
  1-playground.png
- ✓ **M3 Live issue flow**
  docs/flows.md
- ✓ **M4 Present and verify**
  interop-report/live/matrix.md
- ✓ **M5 Protocol inspector**
  2-timeline-and-inspector.png
- ✓ **M6 Two formats**
  interop-report/live/matrix.md
- ✓ **M7 Report per run**
  3-report.png

Good-to-have

- ✓ **G1 Non-Inji mock roles**
  interop-report/reference/matrix.md
- − **G2 mDoc third format**
  interop-report/reference/matrix.md
- ✓ **G3 Sequence diagram**
  9-playback.png
- ✓ **G4 Replay and tamper**
  4-tamper-result.png
- ✓ **G5 History with filters**
  6-history.png
- ✓ **G6 Same-device, DC API**
  13-dc-api.png

Bonus

- − **B1 Nightly matrix**
  .github/workflows/interop-nightly.yml
- − **B2 Gaps as issues**
  docs/gaps/issues/
- ✓ **B3 Telemetry**
  docs/telemetry-obsrv.md
- ✓ **B4 One-command bundle**
  docs/setup.md

Tick: done, with the file shown. Dash: built, one part still open (see Scope). Screenshots are in docs/ux/screenshots/, other paths under docs/. Source: docs/STATUS.md.

IMPACT

149 matrix cells checked, 29 Inji gaps found

Who benefits

Integrators
see compatibility before they commit

Inji teams
get gap reports with issue drafts

ID programmes
compare vendors in one matrix

What changes, counted

149

matrix cells checked

123 reference, 16 live-core, 10 Inji Web

29

interoperability gaps found

in Inji Certify, Verify and Mimoto

27

issue drafts written

2 more reported privately

249 s

npm run demo -- --ci to DEMO PASS
including the image build

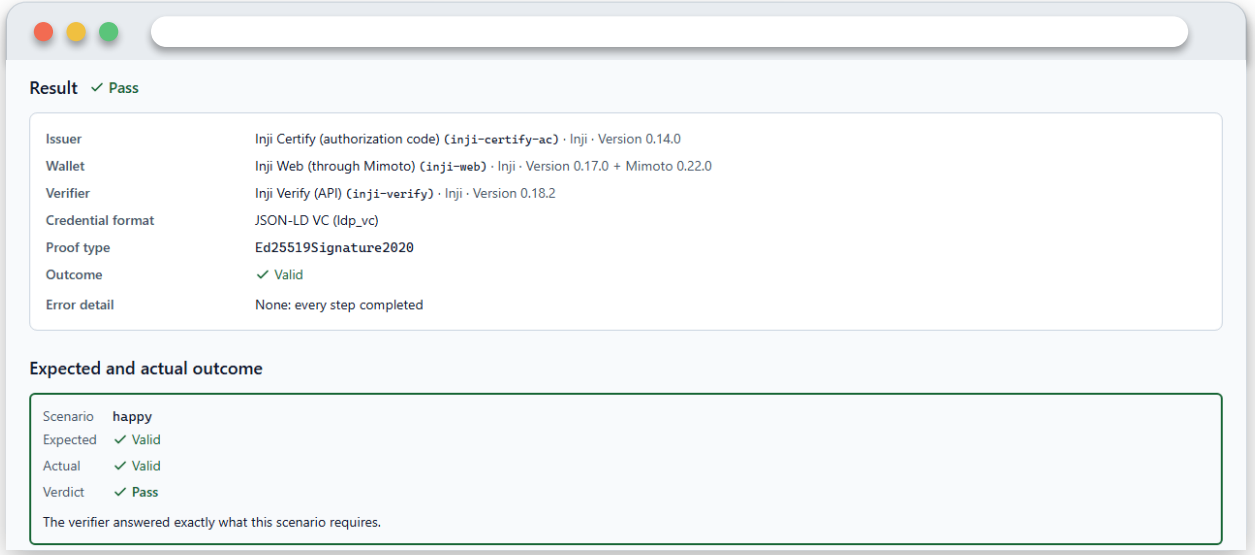
Inji is tested against independent code, in both directions

1. **Inji tested against independent code**
   reference roles written from the specs

2. **Every message, both directions**
   three capture points; exports are redacted

3. **Each tamper case cites its spec text**
   why the verifier must refuse

4. **Beyond the brief**
   Digital Credentials API, mdoc, playback, nightly matrix



Run report: expected and actual outcome per scenario

Scope and next steps

Scope

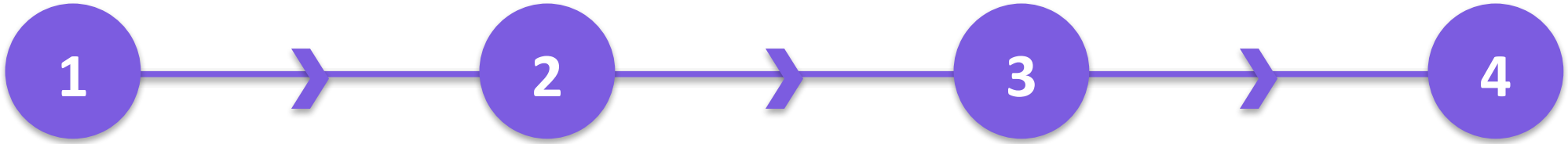
- Inji Wallet mobile not run: no phone; the emulator stays blank
- No mdoc status check; revocation not tested against Inji Verify
- The API on port 28100 has no login; it listens on 127.0.0.1

Runs still to do

- 27 issue drafts not filed upstream yet
- Nightly matrix not yet run against MOSIP Collab
- Obsrv events not tested against a real Obsrv

ROADMAP

Next: file the issue drafts, then run Inji Wallet on a phone



File the issue drafts

G7 and G20 privately

Run the nightly matrix against the Collab sandbox

through repository variables

Run Inji Wallet mobile on a phone

docs/phone-wallet-guide.md

mdoc status check

and mdoc over the Digital Credentials API

TEAM

Team Badmosip

KB

Kanav Bhardwaj

IB

Ivan Bhargava

NM

Nikunj Mahajan

IS

Ishaan Sodhi

HM

Harsh Modi

Repository

https://github.com/Paradox-73/Interoperability_Playground_for_Verifiable_Credentials

Public on GitHub, with the code, the test reports and the screenshots behind every number in this deck

TRY IT

Run it yourself

```
$ npm ci
$ npm run demo
```

You need Docker Desktop (or Docker Engine with Compose 2.24 or newer), Node.js 24, Git, 6 to 8 GB of memory for Docker.

You see The bundle starts, five live cells run, DEMO PASS prints, and the playground stays up on http://localhost:28100. npm run demo took 384 s from a clean checkout on 2026-10-14.

Scan for the repository

Repository **https://github.com/Paradox-73/Interoperability_Playground_for_Verifiable_Credentials**